

Notes: Aghion, Bloom, Blundell, Griffith, and Howitt (QJE, 2005)

Competition and Innovation: An Inverted-U Relationship

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1 Big picture

- **Question.** What is the relationship between product market competition and innovation? Is it monotone, as much of the earlier theoretical literature suggested, or can it be nonmonotonic?
- **Setting.** The paper combines a U.K. industry panel from 1973–1994 with a step-by-step Schumpeterian model. The empirical innovation measure is citation-weighted patents, and the main competition measure is one minus the Lerner index.
- **Core challenge.** Competition and innovation are jointly endogenous. A positive or negative raw correlation could reflect omitted industry characteristics, reverse causality, or common shocks rather than a causal effect of competition on innovation.
- **Main empirical contribution.** The paper estimates a flexible nonlinear relationship between competition and innovation and shows that the data support an inverted-U shape rather than a purely positive or purely negative line.
- **Main theoretical contribution.** The paper builds a model in which competition has two opposite effects:
 - an **escape-competition effect** that encourages neck-and-neck firms to innovate to get ahead;
 - a **Schumpeterian effect** that discourages laggard firms because competition lowers the rents from catching up.
- **Main empirical design.** The baseline uses Poisson count regressions for citation-weighted patents with industry and year fixed effects. To address endogeneity, the paper instruments competition with major U.K. policy changes: Thatcher privatizations, the EU Single Market Programme, and Monopoly and Merger Commission interventions.
- **Main baseline result.** Competition has an inverted-U relationship with innovation. In the preferred annual specification with year and industry effects, the linear term is positive and the squared term is negative. The peak lies near the median competition level in the sample.
- **Main theory result.** Aggregate innovation is high at intermediate competition because low competition mainly activates the escape-competition effect, while very high competition mainly activates the discouragement of laggards.
- **Additional theory predictions.** The model predicts that competition should increase the average technological distance between leaders and followers, and that the inverted-U should be steeper in more neck-and-neck industries.
- **Main supporting evidence.** Both extra predictions are supported in the data: technology gaps are larger in more competitive industries, and the competition–innovation curve is steeper in industries that are more neck-and-neck.
- **Economic takeaway.** Competition can stimulate innovation, but not always. It helps most when firms are close enough technologically that innovating allows them to escape direct competition. When firms are already far behind, stronger competition mainly cuts post-innovation rents and reduces innovative effort.

2 Data and measurement (key objects)

2.1 Data source and sample construction

- The firm-level accounting data come from **Datastream** for firms listed on the London Stock Exchange.
- Patent data come from the **NBER patent database**, matched to the U.K. firms.
- The final firm-level sample contains **311 firms in 17 two-digit SIC industries** over **1973–1994**.
- The main estimating sample is an **unbalanced industry-year panel with 354 observations**.

Interpretation. The paper is not using a cross-section of firms at one date. It uses a long industry panel, which is crucial because the identifying variation comes from changes in competition and innovation within industries over time.

2.2 Why the U.K. is useful in this paper

- The U.K. experienced large and differential product-market reforms across industries in the 1980s and early 1990s.
- These reforms include privatizations, the Single Market Programme, and regulatory interventions by the Monopoly and Merger Commission.
- Those reforms create cross-industry and over-time variation in competition that the paper later exploits for identification.

Interpretation. The U.K. panel is attractive not just because good firm and patent data exist, but because there are policy shocks that plausibly move competition for reasons outside contemporaneous industry innovation.

2.3 Innovation measure

The paper measures innovation with **citation-weighted patents**. Let patents be counted at the industry-year level. Each patent is weighted by the number of later citations it receives.

- Ordinary patent counts treat all patents as equally valuable.
- Citation weighting gives more importance to patents that later receive more attention from other patents.
- The paper uses the *average* citation-weighted patents of firms within an industry-year.

Interpretation. Citation-weighted patents are meant to proxy for innovative output rather than innovative input. The paper chooses them partly because R&D expenditure is poorly reported in the U.K. before 1990.

2.4 Competition measure

The firm-level Lerner-style margin is

$$\ell_{iit} = \frac{\text{operating profit} - \text{financial cost}}{\text{sales}}. \quad (1)$$

The industry-level competition measure is then

$$c_{jt} = 1 - \frac{1}{N_{jt}} \sum_{i \in j} \ell_{iit}, \quad (2)$$

where higher values of c_{jt} imply stronger product market competition.

- A value near one corresponds to price close to marginal cost.
- Lower values indicate more market power.
- The measure is averaged across all listed firms in the industry, not only those in the patenting sample.

Interpretation. The paper uses one minus the Lerner index so that a higher number means *more* competition. That makes the empirical relationship easier to read because a positive coefficient means competition raises innovation locally.

2.5 Technology gap and neck-and-neckness

To test the model's extra predictions, the paper constructs a technology-gap measure from firm-level total factor productivity:

$$m_{it} = \frac{TFP_{Ft} - TFP_{it}}{TFP_{Ft}}, \quad (3)$$

where F denotes the frontier firm in the industry-year.

The industry-level measure m_{jt} is the average of m_{it} across firms in industry j at time t .

- Low m_{jt} means firms are **more neck-and-neck**.
- High m_{jt} means leaders are farther ahead of followers.

Interpretation. This measure is the empirical bridge between the theory and the data. The theory is about the distribution of technological gaps; the empirical proxy uses distance to the TFP frontier.

2.6 Policy instruments

The paper instruments competition using three sets of policy variables:

- **Single Market Programme** exposure in 1988;
- **privatization** episodes under the Thatcher government;
- **Monopoly and Merger Commission** actions that imposed remedies on particular industries.

Industries directly affected include brewing, cars, car parts, periodicals, razors and blades, ordnance, steel, telecoms, and textiles.

Interpretation. These instruments are designed to shift product market competition at the industry level without being chosen in response to short-run within-industry patenting outcomes.

2.7 Descriptive facts

The main descriptive facts from the paper are:

- mean patents: 6.59;
- mean citation-weighted patents: 6.65;
- mean competition measure ($1 - \text{Lerner}$): 0.95;
- mean technology gap: 0.49;
- competition ranges from about 0.87 to 0.99;
- the industry-year panel is fairly small in dimension, which is one reason the paper uses a parsimonious empirical setup.

Interpretation. The mean competition level is already quite high in absolute terms, so the paper is really identifying how changes inside this narrow but economically important range affect innovation.

3 Models and empirical specifications (all equations + interpretation)

3.1 Empirical competition–innovation relationship

The paper starts from a flexible reduced-form relationship between the innovation hazard n and competition c :

$$n = e^{g(c)}, \tag{4}$$

where $g(\cdot)$ is initially left unspecified.

If patent counts follow a Poisson process, then the first moment is

$$E[p \mid c] = e^{g(c)}. \tag{5}$$

Adding controls and fixed effects gives

$$E[p_{jt} \mid c_{jt}, x_{jt}] = \exp(g(c_{jt}) + x'_{jt}\beta), \tag{6}$$

where x_{jt} includes year and industry dummies.

Interpretation. The Poisson structure is used as a count-data workhorse. The key empirical question is the shape of $g(c)$: linear, decreasing, or inverted-U.

3.2 Baseline exponential quadratic specification

The semiparametric spline suggests an inverted-U, and the paper approximates it with an exponential quadratic:

$$E[p_{jt} \mid c_{jt}, x_{jt}] = \exp(\beta_1 c_{jt} + \beta_2 c_{jt}^2 + x'_{jt} \gamma). \quad (7)$$

Main coefficient estimates from Table I are:

- **Column (2), preferred annual specification with year and industry effects:**

$$\hat{\beta}_1 = 387.46, \quad \hat{\beta}_2 = -204.55.$$

- **Column (4), instrumented/control-function version:**

$$\hat{\beta}_1 = 385.13, \quad \hat{\beta}_2 = -204.83.$$

Interpretation. The positive linear term and negative squared term imply that innovation first rises with competition and then falls. The similarity of columns (2) and (4) is one of the strongest empirical messages of the paper: correcting for endogeneity does not materially alter the shape.

3.3 Estimation choice

- The outcome is count-like, so the paper uses **Poisson regressions** based on the first moment.
- The preferred annual specification includes **year fixed effects** and **industry fixed effects**.
- A five-year-average version is estimated as a robustness check to allow for slower adjustment.

Interpretation. The paper is not relying on a purely parametric nonlinear least-squares fit. It embeds the inverted-U in a count-data panel framework with strong fixed effects.

3.4 Endogeneity and the control-function specification

To address endogeneity of competition, the paper specifies a reduced form for competition:

$$c_{jt} = \pi(z_{jt}) + x'_{jt} \zeta + v_{jt}, \quad (8)$$

where z_{jt} are the policy instruments.

The control-function restriction is

$$E[e^{u_{jt}} \mid c_{jt}, x_{jt}, v_{jt}] = 1. \quad (9)$$

- The residual v_{jt} from the competition equation is used as a control in the innovation equation.
- The paper also includes additional foreign-industry variables in the reduced form: import intensity, TFP, operating surplus, and markup measures from France and the United States.

Interpretation. Rather than a textbook two-stage least squares, the paper uses a control-function approach adapted to the nonlinear Poisson count setting.

3.5 Theory: production, technology, and profits

The model has a unit mass of consumers with logarithmic utility over final output:

$$u(y_t) = \ln y_t. \quad (10)$$

Final output is produced from a continuum of intermediate sectors:

$$\ln y_t = \int_0^1 \ln x_{jt} dj, \quad (11)$$

with sectoral output composed of the two firms' goods:

$$x_j = x_{Aj} + x_{Bj}. \quad (12)$$

Firm $i \in \{A, B\}$ with technology level k_i produces according to

$$A_i = \gamma^{k_i}, \quad (13)$$

so one unit of labor generates γ^{k_i} units of output. Equivalently, marginal labor cost is γ^{-k_i} .

- The technology gap m measures how many steps the leader is ahead of the follower.
- For tractability, the paper assumes the maximum sustainable gap is one step: $m \in \{0, 1\}$.

Profits in an unleveled industry are

$$\pi_{-1} = 0, \quad \pi_1 = 1 - \gamma^{-1}. \quad (14)$$

In a neck-and-neck industry, the current profit is parameterized as

$$\pi_0 = \varepsilon \pi_1, \quad 0 \leq \varepsilon \leq \frac{1}{2}, \quad (15)$$

and competition is summarized by

$$\Delta = 1 - \varepsilon. \quad (16)$$

Interpretation. Higher Δ means stronger product market competition. In the model, it lowers the profits of firms that are neck-and-neck before they innovate, which raises the gain from escaping competition.

3.6 R&D technology and Bellman system

If a firm innovates with hazard rate n , it pays R&D cost

$$\psi(n) = \frac{n^2}{2}. \quad (17)$$

A laggard can also catch up mechanically with hazard h even if it does no R&D. Let

- n_0 be R&D intensity for a neck-and-neck firm,

- n_{-1} be R&D intensity for a laggard in an unleveled sector,
- $n_1 = 0$ for the leader in an unleveled sector.

Let V_1 be the value of being a leader one step ahead, V_{-1} the value of being a laggard, and V_0 the value in a neck-and-neck sector. The Bellman equations are

$$rV_1 = \pi_1 + (n_{-1} + h)(V_0 - V_1), \quad (18)$$

$$rV_{-1} = \pi_{-1} + (n_{-1} + h)(V_0 - V_{-1}) - \frac{n_{-1}^2}{2}, \quad (19)$$

$$rV_0 = \pi_0 + n_0(V_1 - V_0) + n_0(V_{-1} - V_0) - \frac{n_0^2}{2}. \quad (20)$$

The first-order conditions are

$$n_{-1} = V_0 - V_{-1}, \quad n_0 = V_1 - V_0. \quad (21)$$

Interpretation. A neck-and-neck firm's innovation effort depends on the gain from jumping ahead. A laggard's effort depends on the gain from catching up. Competition affects those two gains differently.

3.7 Reduced-form R&D equations and Proposition 1

Eliminating the value functions yields

$$\frac{n_0^2}{2} + (r + h)n_0 - (\pi_1 - \pi_0) = 0, \quad (22)$$

$$\frac{n_{-1}^2}{2} + (r + h + n_0)n_{-1} - \pi_0 - \frac{n_0^2}{2} = 0. \quad (23)$$

For the special case $r = 0$, these simplify to

$$n_0 = -h + \sqrt{h^2 + 2\Delta\pi_1}, \quad (24)$$

$$n_{-1} = -(h + n_0) + \sqrt{h^2 + n_0^2 + 2\pi_1}. \quad (25)$$

The paper's **Proposition 1** states:

- n_0 **increases** with competition Δ (escape-competition effect);
- n_{-1} **decreases** with competition Δ (Schumpeterian effect).

Interpretation. Competition is good for innovation when firms are neck-and-neck, because it hurts their pre-innovation profits more than their post-innovation profits. But it is bad for laggards, because it mainly lowers the rents they get if they catch up.

3.8 Steady state, aggregate innovation, and Proposition 2

Let μ_1 be the probability that an industry is unleveled and μ_0 the probability that it is neck-and-neck. Steady-state flow balance requires

$$\mu_1(n_{-1} + h) = 2\mu_0 n_0. \quad (26)$$

Using $\mu_0 + \mu_1 = 1$, aggregate innovation is

$$I = 2\mu_0 n_0 + \mu_1(n_{-1} + h) = \frac{4n_0(n_{-1} + h)}{2n_0 + n_{-1} + h}. \quad (27)$$

Proposition 2 then shows that aggregate innovation follows an inverted-U under economically relevant parameter values.

Interpretation. The inverted-U is a *composition result*. Competition changes not only innovation incentives inside each type of sector, but also the share of time the economy spends in neck-and-neck versus unleveled states.

3.9 Mapping theory to data and additional predictions

The model-implied empirical competition measure is

$$c = 1 - \mu_1 \frac{\pi_1}{2} - 2(1 - \mu_1)(1 - \Delta)\pi_1. \quad (28)$$

Proposition 3 states that this empirical competition measure is increasing in the theoretical competition parameter Δ .

The two main additional predictions are:

- **Proposition 4.** The expected technological gap increases with competition.
- **Proposition 5.** The inverted-U is steeper, and peaks at a higher competition level, in more neck-and-neck industries.

The empirical proxy for technological spread is the industry-average TFP distance to the frontier, m_{jt} , defined earlier.

Interpretation. These extra predictions are valuable because they make the model more than a story that merely rationalizes one curve. The empirical section can actually challenge the model along two additional dimensions.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline estimate

The baseline competition effect is identified from within-industry changes over time in competition and innovation, after removing common year shocks and permanent industry differences.

- **Industry fixed effects** absorb long-run differences in technological opportunity or patenting intensity.
- **Year fixed effects** absorb macro shocks, common patenting trends, and aggregate business-cycle conditions.

Interpretation. Without fixed effects, more innovative industries might also simply be more competitive for reasons unrelated to a causal competition effect. The paper treats the industry panel structure as essential rather than auxiliary.

4.2 Why endogeneity is the main empirical concern

Competition and innovation are mutually endogenous.

- More innovation may itself change market structure and markup behavior.
- Industry-specific demand or technology shocks may move both competition and patenting.
- Omitted factors such as import pressure or productivity shocks may bias a simple regression.

Interpretation. The authors explicitly state that failing to address endogeneity would likely bias estimates toward finding too negative a competition–innovation slope if innovation tends to lower markups.

4.3 Why the policy instruments are plausible

The paper argues that the policy instruments are driven mainly by:

- political ideology and privatization policy,
- European integration through the Single Market Programme,
- and regulatory responses to anticompetitive behavior.

These drivers are not obviously linked to short-run within-industry patenting performance.

Table I also reports supporting first-stage evidence:

- significance of policy instruments in reduced form: 10.11 with $p = 0.002$;
- significance of other instruments in reduced form: 5.00 with $p = 0.000$;
- reduced-form $R^2 = 0.801$.

Interpretation. The paper does not present a modern weak-instrument diagnostic table, but the control-function column is supported by a strong reduced form and by economically coherent policy shifts.

4.4 Robustness logic in the empirical section

The core inverted-U survives several additional checks:

- five-year averages instead of annual data;

- semiparametric spline instead of only the quadratic approximation;
- alternative innovation measure using R&D expenditure, though statistical precision falls because the sample is much smaller;
- top innovating industries estimated separately.

Interpretation. The idea is not that every single specification is equally strong. Rather, many different ways of looking at the data point to the same qualitative shape.

4.5 How the theory helps identification conceptually

The theory sharpens the interpretation of the empirical inverted-U because it explains three things at once:

- why the reduced-form competition effect can be positive at low levels of competition;
- why it can turn negative at high competition;
- and why technological spread should increase with competition.

Interpretation. The extra technological-gap tests are not just decorative. They serve as overidentifying-style evidence for the escape-competition mechanism.

4.6 What the paper can and cannot distinguish

- The paper identifies a robust nonlinear relationship between competition and innovation.
- It also provides a theory that fits the nonlinear pattern and matches additional facts about technological spread.
- But it does *not* separately identify every microchannel through which competition matters; for example, it does not observe firms' R&D projects directly or isolate all organizational channels from the product-market channel.
- The empirical model is also at the industry level, so it cannot describe within-industry firm-to-firm strategic interactions with full richness.

Interpretation. The paper is strongest as a combined empirical-theoretical argument. It is less a pure structural estimation than a disciplined reduced-form paper backed by a mechanism model.

4.7 Relation to the literature

- Relative to traditional Schumpeterian and endogenous-growth models, the paper shows that stronger competition need not always reduce innovation.
- Relative to agency-based models in the spirit of Hart (1983) and Schmidt (1997), the paper gives a product-market mechanism that can generate a full inverted-U rather than a monotone result.
- Relative to Nickell (1996) and Blundell, Griffith, and Van Reenen (1999), it shows why linear empirical specifications may miss the central nonlinearity in the data.

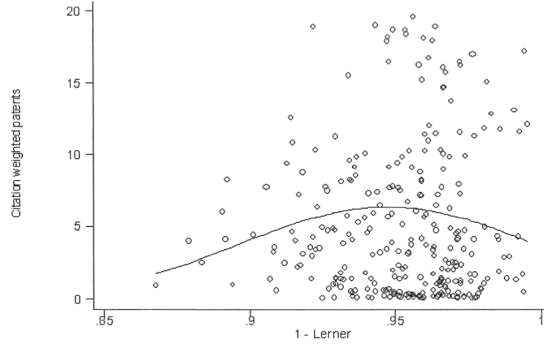


FIGURE I
Scatter Plot of Innovation on Competition
The figure plots a measure of competition on the x -axis against citation-weighted patents on the y -axis. Each point represents an industry-year. The scatter shows all data points that lie in between the tenth and ninetieth deciles in the citation-weighted patents distribution. The exponential quadratic curve that is overlaid is reported in column (2) of Table I.

Interpretation. This paper’s lasting importance is that it changed the benchmark view from “competition is good” or “competition is bad” to “competition can be good up to a point, and the point depends on technological neck-and-neckness.”

5 Tables and figures

5.1 Figure I: scatter plot of innovation on competition

Read:

- The horizontal axis is $1 - \text{Lerner}$, so moving right means more competition.
- The vertical axis is citation-weighted patents.
- The scatter is noisy, but the overlaid quadratic fit clearly bends upward first and downward later.
- The peak appears around competition levels close to 0.95, which is also near the sample median.

Economic message. The raw data already suggest that the linear positive relation found in some earlier work is incomplete. The empirical question is not whether competition is good or bad on average; it is whether the slope changes sign.

5.2 Figure II and Table I: the baseline inverted-U and the instrumented check

Read:

- Figure II overlays the fitted exponential quadratic and a semiparametric spline. The two curves track each other closely.
- Table I, column (2), is the preferred annual specification with year and industry effects. The linear term is positive (387.46) and the squared term is negative (-204.55), with a joint significance statistic of 38.34.
- Table I, column (3), uses five-year averages and preserves the inverted-U.

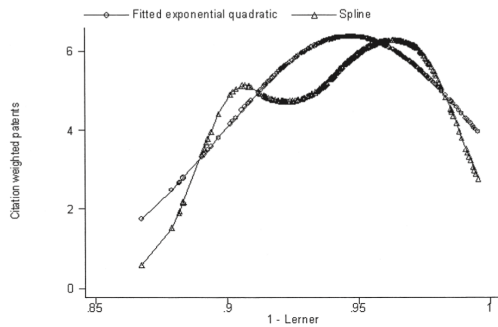


FIGURE II

Innovation and Competition: Exponential Quadratic and the Semiparametric Specifications with Year and Industry Effects

The figure plots a measure of competition on the x-axis against citation-weighted patents on the y-axis. Each point represents an industry-year. The circles show the exponential quadratic curve that is reported in column (2) of Table I. The triangles show a nonparametric spline.

TABLE I
EXPONENTIAL QUADRATIC: BASIC SPECIFICATION

Dependent variable: citation-weighted patents	(1)	(2)	(3)	(4)
Data frequency	Annual	Annual	5-year averages	Annual
Competition _{it}	152.80 (55.74)	387.46 (67.74)	819.44 (265.63)	385.13 (67.56)
Competition squared _{it}	-80.99 (29.61)	-204.55 (36.17)	-434.43 (141.43)	-204.83 (36.06)
Significance of: Competition _{it}	7.60 (0.02)	38.34 (0.00)	9.97 (0.01)	32.59 (0.00)
Competition squared _{it}				10.11 (0.002)
Significance of policy instruments in reduced form				5.00 (0.000)
Significance of other instruments in reduced form				4.38 (0.04)
Control functions in regression				0.801 (0.001)
R ² of reduced form				
Year effects	Yes	Yes	Yes	Yes
Industry effects		Yes	Yes	Yes
Observations	354	354	67	354

Competition_{it} is measured by (1-Lerner index) in the industry-year. All columns are estimated using an unbalanced panel of seventeen industries over the period 1973 to 1994. Estimates are from a Poisson regression. Numbers in brackets are standard errors. The standard errors in column (4) have not been corrected for the inclusion of the control function. Significance tests show likelihood ratio test-statistics and *P*-value from the *F*-test of joint significance. The fourth column includes a control function. The excluded variables are policy instruments specified in Table II, imports over value-added in the same industry-year, TFP in the same industry-year, output minus variable costs over output in the same industry-year and estimates of markups from industry-country regression [Martins et al. 1996] interacted with time trend, all for the United States and France.

TABLE II
POLICY INSTRUMENTS

Industry	Year(s)	Policy
All, but differential impacts	1988	Single Market Program
Brewing	1986	MMC action
Cars	1984, 1987, 1988	MMC action, Privatization
Car parts	1982, 1987	MMC action, Privatization
Periodicals	1987	MMC action
Razors and blades	1990	MMC action
Ordnance	1987	Privatization
Steel	1987	Privatization
Telecoms	1981, 1984, 1989	MMC action, Privatization
Textiles	1989	MMC action

- Table I, column (4), uses the policy-instrument control function. The coefficients remain very close to the baseline: 385.13 on competition and -204.83 on competition squared.

Economic message. The inverted-U is not a fragile artifact of the exact functional form. The spline confirms the shape, and the control-function column shows that the curve survives the main endogeneity correction.

5.3 Table II: policy instruments

Read:

- The table lists the policy shocks used to identify competition, including the Single Market Programme in 1988 and several industry-specific privatization and MMC actions.
- Telecoms, cars, car parts, ordnance, and steel were affected by privatizations; brewing, periodicals, razors and blades, and textiles were affected by MMC actions.
- The policy timing varies by industry, which is exactly what the panel design needs.

Economic message. The paper's identification is not coming from one single reform. It comes from a set of staggered policy shocks that move competition differently across industries and years.

5.4 Table III and Figure III: technological spread and the neck-and-neck split

TABLE III
TECHNOLOGY GAP AND EXPONENTIAL QUADRATIC WITH NECK-AND-NECK SPLIT

Dependent variable:	(1) Technology gap	(2) Technology gap	(3) Citation-weighted patents	(4) Citation-weighted patents
Estimation procedure	Linear regression	Linear regression	Poisson	Poisson
Competition _{it}	2.858 (0.400)	0.942 (0.419)	183.81 (58.99)	424.46 (69.5)
Competition squared _{it}			-96.35 (31.01)	-222.9 (36.9)
Competition _{it} * Technology gap _{it}			1.43 (2.48)	3.82 (2.66)
Competition squared _{it} * Technology gap _{it}			-1.30 (2.59)	-3.84 (2.78)
Significance of:				
Competition _{it}			16.59 (0.00)	39.21 (0.00)
Competition squared _{it}				
Significance of:				
Competition _{it} * Technology gap _{it}			9.74	7.93
Competition squared _{it} * Technology gap _{it}			(0.01)	(0.02)
Year effects	Yes	Yes	Yes	Yes
Industry effects		Yes		Yes

Competition_{it} is measured by (1-Lerner index) in the industry-year. Technology gap_{it} is measured by the average distance to the TFP frontier firm across all firms in the industry-year, so it is an inverse measure of neck-and-neckness. All columns estimated using an unbalanced panel of 354 yearly observations on seventeen industries over the period 1973 to 1994. Significance tests show likelihood ratio test-statistics and *P*-value from the *F*-test of joint significance. Numbers in brackets are standard errors. The standard errors in columns 3 and 4 have not been corrected for the inclusion of the control function.

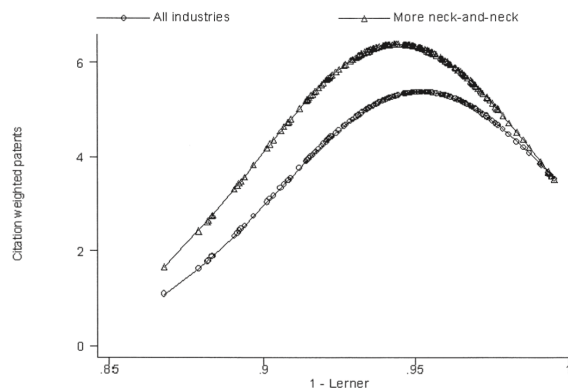


FIGURE III
Innovation and Competition: The Neck-and-Neck Split
The figure plots a measure of competition on the x-axis against citation-weighted patents on the y-axis. Each point represents an industry-year. The circles show the exponential quadratic curve that is reported in column (2) of Table I. The triangles show the exponential quadratic curve estimated only on neck-and-neck industries that is reported in column (4) of Table III.

Read:

- Table III, columns (1) and (2), regress the technology gap on competition. The coefficients are positive: 2.858 without industry fixed effects and 0.942 with them.
- Table III, columns (3) and (4), interact competition with the technology gap in the patent equation. The interaction terms make the inverted-U steeper in more neck-and-neck industries.
- Figure III shows exactly this graphically: the curve for more neck-and-neck industries sits above the baseline and is more sharply curved.

Economic message. These are the paper's most important corroborating tests. Competition increases technological spread, and the escape-competition effect is stronger when firms begin close to one another.

5.5 Table IV: descriptive statistics

Read:

- Citation-weighted patents are highly skewed: the mean is 6.65, the median is 3.35, and many industry-years have zero patents.
- Competition (1 - Lerner) is concentrated in a fairly tight band, with mean and median both around 0.95.
- The technology gap has mean 0.49 and median 0.51, indicating substantial within-industry productivity dispersion.

TABLE IV
DESCRIPTIVE STATISTICS

	Mean (s.d.)	Median	Min	Max
Patents	6.59 (8.52)	3.5	0	54
Cite weighted patents	6.65 (8.43)	3.35	0	45
1-Lerner	0.95 (0.023)	0.95	0.87	0.99
Technology gap (m)	0.49 (0.155)	0.51	0.080	0.81

The sample is an unbalanced panel of 354 yearly observations on seventeen industries over the period 1973 to 1994.

Economic message. The descriptive table makes clear why nonlinearities matter. Innovation is sparse and skewed, while competition varies over a narrow interval where a change in slope is economically possible.

6 Short summary

- The paper studies the effect of product market competition on innovation using U.K. industry data and a model of step-by-step innovation.
- Empirically, competition and innovation are related by an inverted-U: innovation first rises and then falls as competition increases.
- The nonlinear pattern survives fixed effects, semiparametric estimation, five-year averaging, and a policy-based endogeneity correction.
- The theory explains the inverted-U with two opposing forces: an escape-competition effect for neck-and-neck firms and a Schumpeterian discouragement effect for laggards.
- The model predicts that competition widens technological gaps and that the inverted-U is steeper in more neck-and-neck industries.
- Both additional predictions are supported in the data.
- The paper's key lesson is that competition can encourage innovation, but mainly when firms are close enough that innovating lets them escape immediate product-market rivalry.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that the competition–innovation relationship is not well described by a straight line. In the U.K. data it is inverted-U shaped, and a Schumpeterian step-by-step model explains why: competition stimulates innovation among neck-and-neck firms but discourages innovation by laggards. The empirical evidence on technology gaps and neck-and-neck industries lines up closely with this mechanism.

7.2 Economic intuition

Competition affects pre-innovation and post-innovation rents differently. When firms are close competitors, stronger competition hurts them badly in the status quo, so getting ahead becomes more attractive. When firms are already behind, stronger competition mainly reduces the rents from catching up, so innovation becomes less attractive. Aggregate innovation is highest in the middle because industry composition shifts endogenously between those two cases.

7.3 Takeaway

This paper's big message is that more competition is not automatically better or worse for innovation. The answer depends on how close firms are technologically and on whether competition mainly creates an incentive to escape or mainly cuts the reward from catching up.